

Ivan Lau | Curriculum Vitae

National University of Singapore, School of Computing

🏠 <https://ivanphlau.github.io/>

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CURRENT RESEARCH INTERESTS

Sequential decision-making under uncertainty and/or communication constraints

EDUCATION

Ph.D. in Computer Science

August 2024 - Present

National University of Singapore

Research topics: Sequential Learning Problems with 1-bit Feedback

CGPA: 5.0/5.0

Advisor: Jonathan Scarlett

Graduate student in Electrical and Computer Engineering

August 2022 - December 2023

Rice University

Research topics: Decentralized Optimization

CGPA: 4.00/4.00

Advisors: César A. Uribe and Shiqian Ma

M.Sc. in Mathematics

September 2019 - August 2021

Simon Fraser University

CGPA: 4.07/4.33

Thesis: 🌀 Nonuniform Compressed Sensing Schemes with Sublinear Measurements, Sublinear Time, and Low Entropy (awarded Certificate with Distinction)

Advisor: Jonathan Jedwab

B.Sc. (Hons) in Computer Science and Mathematics

September 2015 - June 2019

University of Edinburgh

Degree classification: First-Class Honours

Mathematics project: 🌀 Left Braces and the Solutions of the Yang-Baxter Equation (awarded best project)

Informatics project: 🌀 Hermitian Spectral Theory of Mixed Graphs

EMPLOYMENT

National University of Singapore

January 2024 - August 2024

Research Assistant, Department of Computer Science

Rice University

May 2023 - December 2023

Research Assistant, Department of Electrical and Computer Engineering

National University of Singapore

September 2021 - August 2022

Research Assistant, Department of Computer Science

Simon Fraser University
Research Assistant, Department of Mathematics

September 2019 - August 2021

University of Edinburgh
Teaching Assistant, School of Informatics

September 2017 - May 2019

University of Edinburgh
Research Intern, Laboratory for Foundations of Computer Science

June 2018 - August 2018

PUBLICATIONS

Journal Papers

8. Model-Based and Graph-Based Priors for Group Testing

Ivan Lau, Jonathan Scarlett, Yang Sun
IEEE Transactions on Signal Processing, 2022.

7. An Associative Left Brace is a Ring

Ivan Lau
Journal of Algebra and Its Applications, 2020

Machine Learning Conference Papers

6. Open Problem: Is Interaction Necessary for Order-Optimal 1-bit Mean Estimation?

Ivan Lau, Jonathan Scarlett
Conference on Learning Theory (COLT) 2026.

5. Sequential 1-bit Mean Estimation with Near-Optimal Sample Complexity

Ivan Lau, Jonathan Scarlett
International Conference on Artificial Intelligence and Statistics (AISTATS) 2026.

4. Quantile Multi-Armed Bandits with 1-bit Feedback

Ivan Lau, Jonathan Scarlett
International Conference on Algorithmic Learning Theory (ALT), 2025.

3. Max-Quantile Grouped Infinite-Arm Bandits

Ivan Lau, Yan Hao Ling, Mayank Shrivastava, Jonathan Scarlett
International Conference on Algorithmic Learning Theory (ALT), 2023

IEEE Conference Papers

2. Decentralized and Equitable Optimal Transport

Ivan Lau, Shiqian Ma, César A. Uribe
American Control Conference (ACC), 2024

1. Construction of Binary Matrices for Near-optimal Compressed Sensing

Ivan Lau and Jonathan Jedwab
IEEE International Symposium on Information Theory (ISIT), 2021.

PREPRINTS

In Submission

3. Non-Adaptive 1-Bit Mean Estimation: Minimax Rates and the Sample-Interval Tradeoff

Ivan Lau, Jonathan Scarlett

2. Batched Stochastic Linear Bandits with 1-Bit Communication Constraints

Ivan Lau, Daniel McMorro, Kevin Jamieson, Jonathan Scarlett

1. Order-Optimal Sequential 1-Bit Mean Estimation in General Tail Regimes

Ivan Lau, Jonathan Scarlett

TEACHING EXPERIENCE

National University of Singapore

The Algorithm Designer's Toolkit (Teaching Assistant)

Spring 2025

Rice University

Random Signals (Teaching Assistant)

Fall 2023

Simon Fraser University

Calculus Workshop (Teaching Assistant)

Fall 2020

Applied Calculus Workshop (Teaching Assistant)

Summer 2020

Computing with Linear Algebra (Teaching Assistant)

Spring 2020

Algebra Workshop (Teaching Assistant)

Spring 2020, Fall 2019

University of Edinburgh

Informatics 2B - Algorithms, Data Structures, Learning (Teaching Assistant)

Spring 2019

Informatics 2D - Reasoning and Agents (Teaching Assistant)

Spring 2019, Spring 2018

Algorithms and Data Structures (Teaching Assistant)

Spring 2019

Discrete Mathematics and Mathematical Reasoning (Teaching Assistant)

Fall 2018, Fall 2017

Informatics 1 - Cognitive Science (Teaching Assistant)

Spring 2018

AWARDS AND HONOURS

National University of Singapore Research Achievement Award

2025/2026

National University of Singapore Research Scholarship

2024

Rice University Department of Electrical and Computer Engineering Fellowship

2022

Simon Fraser University Department of Mathematics Certificate "With Distinction"

2021

Simon Fraser University Graduate Fellowship and Special Graduate Entrance Scholarship

2019

University of Edinburgh William and Isabella Dick Fourth Year Project Prize

2019

Malaysian Computing Competition (Perfect Score)

2014, 2013

Malaysian National Mathematical Olympiad (4th - 13th Place Bracket)

2013

International Competitions and Assessments for School (Gold Medal in Science)

2012

PROFESSIONAL ACTIVITIES

Reviewer for IEEE Trans. Inf. Theory

2026 - Present

Reviewer for AISTATS, ICLR, NeurIPS

2025 - Present

Reviewer for ICASSP, ACC

2024 - Present

Skills

Python, R, MATLAB, $\LaTeX$